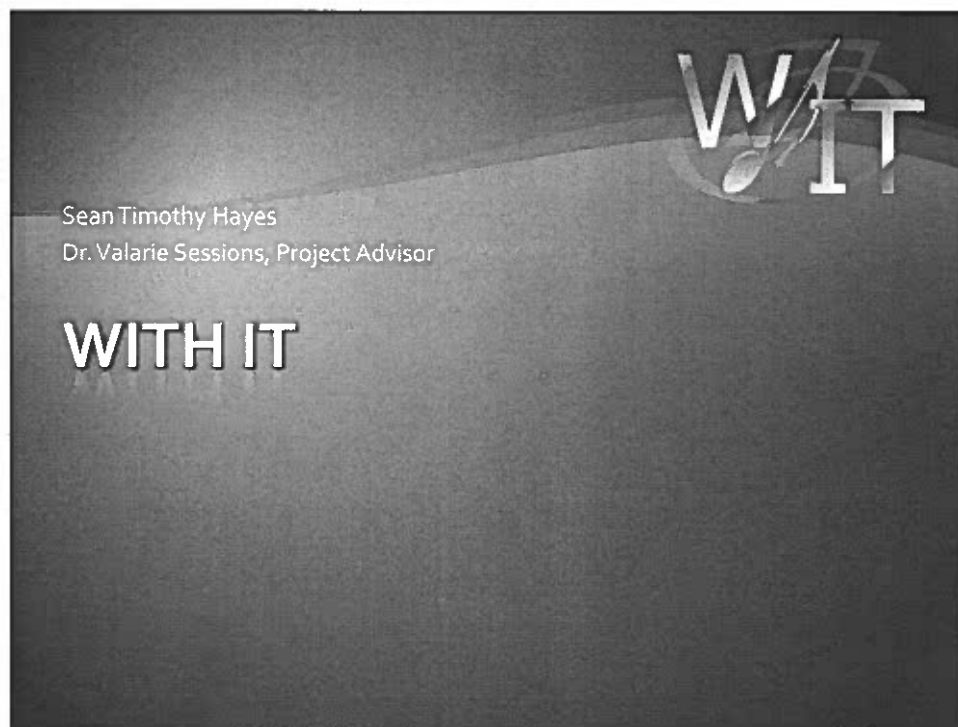






Project Purpose

- Growth as a musician relies on a strong attention to detail
- The repetition necessary for progress may lead to hidden errors.
- *With It* allows musicians to check their accuracy and discover these errors.



Key Elements

- Metronome
- Tuner
- Analyzer





Research

Music and Frequency

- Overtones
- Fast Fourier Transform
- Determining the Fundamental
- ♦ MIDI Protocol



Code

Portability

Libraries

- JMusic
- MARF
- Impro-Visor





Challenges

- Accurate FFT Results
- Determining the Fundamental
 - Method 1: Lowest Frequency with Large Volume
 - Method 2: Greatest Common Factor
 - Method 3: Hybrid Model
- Displaying Music
- Rhythm Analysis



Code Highlights

- Multithreaded
- Pulse Event and Listener
- Basic MIDI Playback

```

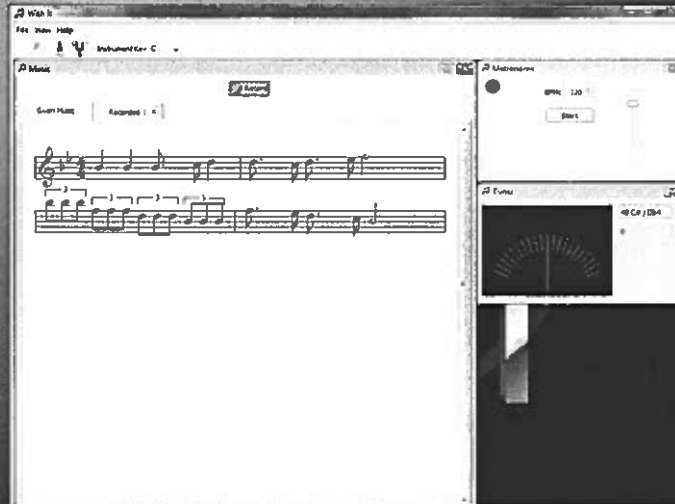
/**
 * This private class is used to fire a Pulse event
 */
private void firePulseEvent()
{
    PulseEvent event = new PulseEvent( this );
    Object[] listeners = listenerList.getListenerList();
    // Each listener occupies two elements - the first is the listener class
    // and the second is the listener instance
    for( int i = 0; i < listeners.length; i += 2 )
    {
        if ( listeners[i] == PulseListener.class )
        {
            ((PulseListener)listeners[i+1]).onPulse( event );
        }
    }
}

```

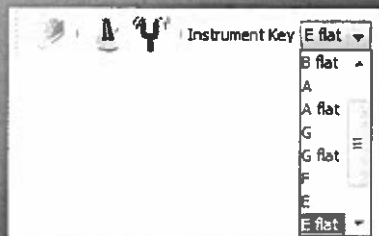
Program Tour



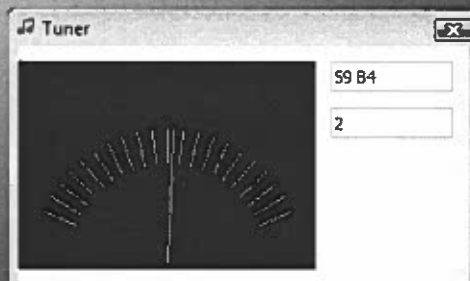
Main Frame



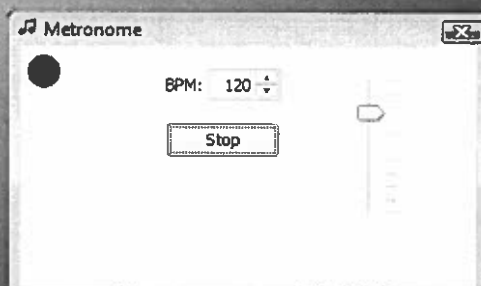
Toolbar



Tuner



Metronome



Analyzer

